

AI System to Automatically Review and Summarize Research Papers – Milestone 4

Statement of Purpose

Milestone 4 represents the final stage of the project where the system is refined, evaluated, and prepared for demonstration. The primary objective is to ensure high-quality output through an automated review and refinement cycle. This stage focuses on improving clarity, coherence, academic accuracy, and usability. Additionally, a polished and user-friendly interface is implemented to integrate all developed modules into a complete working system ready for final submission.

Scope of Project

The scope of this milestone includes adding automated evaluation mechanisms that assess generated drafts based on clarity, structure, completeness, and academic tone. The system provides revision suggestions and allows iterative improvement. All modules including text extraction, synthesis, and draft generation are integrated into a clean Gradio interface. The final output is compiled into a structured report with proper formatting and ready for presentation.

Detailed Methodology

- Step 1: Implemented an automated review cycle to re-evaluate generated sections and detect areas needing refinement.
- Step 2: Developed a quality evaluation module that scores drafts based on coherence, completeness, and academic writing standards.
- Step 3: Created a revision suggestion system to provide actionable improvements for each section.
- Step 4: Designed and implemented a polished Gradio-based user interface for smooth user interaction.
- Step 5: Added interactive UI controls such as 'Critique/Revise' button to allow user-triggered refinement.
- Step 6: Integrated Abstract, Methods comparison, Results synthesis, and APA references into a consolidated final report.
- Step 7: Conducted final system testing, debugging, documentation, and prepared presentation materials.

Week 7–8 Implementation Tasks

- Implemented iterative review and refinement cycle for quality improvement.
- Integrated revision suggestion and evaluation modules.
- Completed final UI integration using Gradio framework.
- Added user-controlled revision triggers.
- Compiled complete final report with all generated sections.
- Performed final testing, documentation preparation, and presentation setup.

System Flow Explanation

1. Start – The system begins execution.
2. Load Generated Draft Sections – Previously generated sections are loaded into the evaluation module.
3. Quality Evaluation Module – The system analyzes clarity, structure, and completeness.
4. Generate Revision Suggestions – Specific improvement recommendations are produced.
5. User Decision – If the user selects 'Critique/Revise', the refinement cycle re-executes.
6. Integrate Final Sections – Approved sections are displayed in the UI.
7. Prepare Final Report – All sections are compiled into a structured document.
8. Final Testing and Documentation – The system is validated and documented.
9. Stop – Final project version is completed and ready for demonstration.